Quantifying the Impact of Weather on Performance in the National Football League

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Introduction

The National Football League (NFL) is a professional American Football league located in the United States. It is comprised of 32 teams across the continental US. The NFL is the most popular sports league in America and continues to expand its global footprint. According to data published by Global Web Index and Statista, the NFL has approximately 115 million fans domestically and more than 80 million fans internationally (McCarthy, 2015). This sizable following generates an estimated 18 billion dollars annually in revenue (Gough, 2023). As evidenced by the magnitude of the viewership and the rapidly increasing revenues, many parties are interested in the results of the games in each NFL season.

There are many variables that impact win-loss outcomes in an NFL game. For games played outdoors, one of those variables is climate. Organizations can face regional climate challenges or local climate challenges, unique to teams in specific cities. Additionally, the NFL season spans from September to early February, exposing teams to summer, fall and winter elements. The varying geographic regions coupled with seasonal changes over a 5-month span often results in games played in a variety of extreme weather. As a result, weather may impact games in a way that affects all interested parties – league organizations, fans, and betting markets among them.

Problem

Approximately 66% of the league's teams play in geographic regions or cities prone to extreme heat, strong wind, heavy precipitation, harsh cold or a combination of these conditions. Games played in these environments have added variance, resulting in ambiguity in the outcome of a game. This uncertainty creates problems for the interested groups outlined above. Within organizations, coaches are required to create game plans best suited for their system, roster strengths and opponent. They may lack a data driven solution to help form a climate specific game plan. Sports betting is seen as a "black box." Sports bettors may have an intuitive understanding of how weather affects betting lines, but many lack specific knowledge of how lines are created and how weather elements may create movement in betting lines. This gap in approach is a limiting factor for sports bettors seeking a betting edge against the sportsbooks. Lastly, many Fantasy Football consumers set lineups based on predicted points and weekly matchups. Missing from their analysis is a way to quantify adverse or advantageous weather conditions and predict statistical output using weather inputs. Our analysis will address the problems facing these three groups and present workable solutions for each.

Literature Survey

In an extensive review of current literature, we learned that multiple studies analyze the effects of weather using game data (Zipperman, 2014; Binney & Smoliga, 2022; Illmer & Daumann, 2022), while other studies used surveys (Sambrook et al., 2023), simulations (Mei & Qu, 2016), or conducted their own research with participants (Calderón-Pellegrino et al., 2020; Mallen et al., 2023; Cooper et al., 2016; Schneider et al., 2024). However, the two common limitations that we found were that professional football players were not used when conducting this research (Illmer & Daumann, 2022; Schneider et al., 2024; Mallen et al., 2023) and the sample size of the testing environment, whether it was the number of athletes (Mallen et al., 2023), stadiums (Mei & Qu, 2016), or the number and types of games (Hallows, 2021; Zipperman, 2014; Cooper et al., 2016; Calderón-Pellegrino et al., 2020), were too small to come to an appropriate conclusion. We also learned that available research provides guidance on health for athletes. Athletes in cold weather should be producing as much heat as they are losing (Gatterer et al., 2021). Athletes suffer negative health outcomes in UV radiation, extreme weather, and heat (Schneider &

Mücke, 2021) and surveyed athletes believe they are negatively affected by extreme heat (Sambrook et al., 2023) and have decreased motivation to exercise at all (Wagner et al., 2019). Measures such as shade, electrolytes and cooperation with weather services have been shown to alleviate some of those negative effects (Schneider, et al., 2024). The current research also shows conflicting results on the impact of humidity on rushing and win-loss outcomes (Paul, 2017; Borghesi, 2008). We plan to build on the existing studies and deviate in new and exciting ways outlined in our proposed method.

Proposed Method

To differentiate our approach from current literature/studies, our analysis will focus on the impact weather has on NFL games for three different demographics: NFL organizations, the sports betting community, and Fantasy Football consumers. While present studies incorporate aspects of weather impact on sports performance and American football, we are not aware of any that attempt to analyze weather and add value for multiple segments of the NFL ecosystem. This study will be unique in totality and in how it dissects weather impact on behalf of each listed demographic.

Our study also took a unique approach to data collection. We utilized nflfastR, an R package that scrapes play-by-play data from NFL.com from 2001 to the most recent season. This gives us access to more than 11 million plays with over 375 variables for each play. Two important data cleaning measures we thought needed further clarification: the exclusion of postseason plays and plays we identified as "garbage time." Postseason plays were excluded due to the lack of relevance for fantasy sports. Garbage time plays are defined as plays that occur where either team has above 95% or below 5% chance of winning (detailed in the win probability variable, wp). While in garbage time, the team that is losing tends to rely heavily on passing plays to try to make up the difference in scoring margin. The team winning skews heavily towards running plays to mitigate risk and expire the clock. While coaches may set aside a handful of plays for garbage time situations, plays called in a neutral game state are more representative of what a coach may game plan for. These data filters are not alone novel but do make our singular analysis more thorough.

Weather-based Game Plan Creation

At the NFL organizational level, one of the most important jobs of the coaching staff for each NFL team is to formulate a game plan for their upcoming opponent. This process begins at the beginning of the game week and is finalized a couple of days before the game (traditionally, NFL games are played on Thursday, Sunday, or Monday). The game plan consists of plays within a coach's offensive system and emphasizes the team's strengths, while capitalizing on opponent weaknesses and favorable matchups. While it is reasonable to assume that coaches consider weather in each game plan, especially in extreme scenarios, there is no publicly available information indicating coaches use data to form weather-based game plans. Our analysis for coaches is a completely innovative concept and contains three parts:

1. Generate a general game plan using expertise from an industry expert and historical NFL play-by-play data from 2001 – 2023. This will serve as the starting point for a weather based gameplan.
2. Utilize regression, variable selection, and random forest to identify the key weather variables that impact offensive success.
3. Apply knowledge from modeling to the original general game plan to output a new game plan, given various weather inputs.

To craft the general game plan, we first examined how game plans are structured. According to J.T. O'Sullivan, a former NFL quarterback with 11 years of experience, NFL game plans are typically broken down into four categories: base runs and base passes (categorized as plays occurring on first down or second down and short or medium yards to go), third downs (short, medium or long yards to go) and red zone (plays inside the opponent's 20 yard line) ("How's an NFL Game Plan Created?," 00:01:07-00:2:00). Understanding this, we utilized play-by-play data to check the frequency with which coaches

called runs plays (left, right, or middle and behind which linemen) and pass plays (deep or short and left, middle, or right) in each of these scenarios.

After bucketing these plays in accordance with the game plan structure definitions, we can see how coaches have utilized game plans to call plays in these scenarios for the last 23 years. A sample of the breakdown can be seen in Appendix A.

With these play call frequencies, we were able to calculate the rates with which coaches have called specific types of plays in a game plan scenario to populate a very general game plan. Our general game plan is about 180 plays, comfortably within the 150-200 play range of a standard NFL game plan ("How's an NFL Game Plan Created?", 00:02:15- 00:2:25). We visualized our game plan as a multi layered, interactive treemap. A visual sample can be seen in Appendix B-1.

To go from the general game plan to the weather driven game plan, we chose to explore the relationship between team success rate and statistically significant weather variables. Success rate is an efficiency metric that "measure[s] the consistency of a team's performance on a play-by-play basis" (Kapadia et al., 2020). Calculating a team's success rate in a game is achieved by dividing the number of successful plays divided by the number of total plays. A successful play in American football is defined as a play that produces positive expected points added (EPA). Each play has a value of expected points, calculated in our dataset using seconds remaining in the half, yard line, possessing team being home or away, roof type, down, yards to go, era, and timeouts remaining for each team (Baldwin, 2021). EPA is the value achieved above (positive) or below (negative) that expectation on a given play. We grouped the play-by-play data by the game id and team with possession variables to create summary game data, each row representing the offensive performance of one team in one game. Each row now summarized each game for each team, producing the sum of successful plays and sum of total plays and allowing us to calculate each team's success rate. Modeling team success rates at the game level would allow examination of weather impact while eliminating some of the noise that would be created by modeling success with weather inputs on a per play basis. Success rate became the dependent variable of the first regression model, with independent variables including the offensive ranking and opponent defensive ranking (on a league ranking basis to best reflect the performance of those units during that season), rain, snow, temperature, wind, humidity, an interaction variable between temperature and humidity, and stadium conditions (indoor/outdoor and grass/turf). Through multicollinearity check and forward selection variable selection, temperature, and humidity (and the interaction variable) were completely removed from our model - a key development for the experimentation and evaluation of our analysis. In our first regression model, we learned that wind, rain and snow were statistically significant weather variables, along with the ranking variables and stadium variables outlined above. Our random forest, weighted methods, and weather game plans will be outlined in our evaluations and conclusions.

Exploring Weather-related Betting Edges

The second innovative way that we are utilizing our predictive model is in the growing market of sports betting. As of February 2024, about half of the states in the U.S. have legalized mobile sports betting (Goldman Sachs, 2024). This quick rise can be attributed to the increased accessibility. There are now several different apps people can download and instantaneously place bets on such as DraftKings, FanDuel, Bet365, and BetMGM. One way that app-based betting has revolutionized the sports betting industry is an increased offering in the types of wagers users can make. Previously, users would mostly place wagers on winners, points spreads and total points. Now, mobile betting has expanded to live betting, allowing users to bet on single play outcomes. Bettors may also build multi-leg parlays that can generate huge payouts comparable to winning the lottery (Goldman Sachs 2024). The accessibility and possibility of winning a large payout make the sports betting industry so attractive to consumers. With betting odds designed to give the sportsbooks the advantage in the long term, bettors would be looking for any edge to tip the scales in their favor.

According to a study conducted by economic researchers, the gambling sector is heavily invested in all factors that can impact performance and scoring, including weather (Borghesi, 2008). The tool generated from our analysis of weather on performance in NFL games could potentially add value to two

parties - sportsbooks/mobile betting companies and sports bettors. However, we focused primarily on how our model can benefit the bettors and give them an edge on their wagers due to the black box nature and multiple factors that contribute to betting line prices. Successfully implementing our model on weather impact on NFL games could provide high returns on investment if our predictors are appropriately understood and utilized properly.

Our predictive model is designed to project numerous statistics from football games including total running yards, total passing yards, and total points scored given a pair of home and away teams for a match. The input data integrates various weather data points such as wind speed, temperature, and humidity, alongside detailed play-by-play game data from the previous 23 years. To uphold the precision and dependability of our predictive tool, we performed k-fold cross-validation for a linear regression model. This approach systematically trains and evaluates the model using several data subsets, improving the model's generalizability and avoids overfitting. Additionally, it facilitates thorough performance evaluation and ensures the model's capability to generalize accurately new, unseen data. Utilizing our predictive tool, a customer can make better informed decisions when it comes to their bets, such as having better predictions on the over/under on the passing yards in a game, when adverse weather conditions such as rain or freezing temperatures are expected.

For this section, our model utilized the Scikit-learn (sklearn) and pandas libraries within a Jupyter Notebook in Pycharm for tasks like data loading, preprocessing, model creation, evaluation, and selection. After spending time to assess, re-assess, and refine the model to enhance its performance, we developed interactive visualizations using Tableau based on the coefficients of our linear model. These visualizations serve as a user interface, taking inputs from users and producing desired game statistics to support their decision-making in betting scenarios. A demo of the interactive UI can be seen in Appendix C.

Projecting Fantasy Football Outcomes with Weather

For Fantasy Football consumers, we created an NFL weather impact prediction tool that can help Fantasy Football managers make data-driven decisions about what players to start in their lineups based on how weather forecasts impact their performance. Each game week, Fantasy Football managers have lineup optionality; they can keep the same lineup as the previous week, swap out usual starters for players on their bench, pickup available players from the league's free agency pool, etc. Traditionally, Fantasy Football managers consider a variety of factors when managing their teams. This can range from general skillset, injuries, off the field issues and violations, etc. The goal for our model is to provide these Fantasy managers with another important consideration to factor that is unique enough to give them an edge over the other managers in their league. If a Fantasy manager knows that the NFL team of their top player will be playing in adverse weather conditions, our model will help them make the decision to play them or not depending on the positive or negative impact that condition has according to this model. It can also assist in position grouping, too. For example, there is typically a Flex position in Fantasy Football rosters where a manager can start an additional player (normally at the running back, wide receiver, or tight end position). With our model, if the specific weather forecast has a negative impact on rushing yards, the Fantasy Football manager would know that it would be more beneficial for them to start either a wide receiver or tight end in that Flex position.

Our inputs for this third predictive model contain multiple NFL statistics from dataset including points scored, rushing yards, and passing yards since 2001. There are also weather-specific statistics such as humidity, temperature, and wind speed. These serve as our independent variables, while the NFL statistics are the dependent variables. Using these metrics, we developed a linear regression model to predict the effect that these weather conditions have on NFL statistics. We utilized a train-test-split methodology for validating our regression model to ensure it has been sufficiently prepared to deliver accurate results. Plus, we have developed an additional interactive visualization to display the results (Appendix D).

Experiments & Evaluation

The main questions that our experiment set out to answer were as follows:

- What game plan style is most effective for certain weather conditions?
- Which weather conditions give bettors a positive Expected Value?
- How does weather affect various positions in Fantasy Football?

After careful consideration, we decided the best methodology for experimenting and evaluating our analysis was utilizing feature subsets with current data, standard train test splits and k-fold cross validation. We considered alternative methods (data from other professional leagues, computer sims, existing large language model sims, and placing and tracking wagers) but decided the best, most scalable solution was in the current data and future iterations of the same dataset.

In our experimentation with linear regression models, we explored various feature combinations and transformations utilizing the play-by-play game data and weather conditions. Additionally, we conducted hyperparameter tuning, focused on optimizing model performance via adjustments to learning rate and regularization strength. Experimentation for the random forest featured in game planning utilized hyperparameter tuning for parameters such as the number of trees and feature subset sizes.

K-fold cross-validation is implemented when evaluating linear regression models to increase the generalizability of the model and avoid overfitting, and we assessed reliability using metrics such as Mean Squared Error (MSE) and R-squared score. In the evaluation of our random forest, percentage of variance explained was the key metric. The modeling utilizing one thousand trees and the subset of variables captured about 19.7% of the variance. As noted, the most impactful predictors of success rate are the team's offense and the opponent's defense. Of our weather variables, wind clearly has the most significant impact on success rate, followed by rain and snow (Appendix E). This is not an unexpected finding, as predicting team performance (in this case, through success rate) is notoriously difficult.

We feel evaluating the success of our tools can best be done using adverse weather games in future seasons as a test bed. Seeing how our game plan tool performs in predicting successful plays can inform us about our methodology and show where improvements can be made. While unlikely, it would be helpful to see the game plan implemented in a real-world scenario and receive feedback from coaches. We can also track betting results aided by our betting tool in extreme weather games. Lastly, we can observe deltas in actual Fantasy performance and predicted Fantasy performance from our regression tool and adjust accordingly.

Some additional observations that we noticed when it came to our game plan model were teams should be passing more in general. Pass plays had a higher success rate in every condition except for snow. Also, running plays are unchanged by weather, but passing plays are negatively impacted by more wind, rain and especially by snow (Appendix F). This information was paramount in applying a weighted method to the number of play values our general game plan to return a weather optimized game plan based on conditions. The weight used for wind was 1% increase and 1% decrease for run plays and pass plays respectively for each mile per hour of wind. Given the drastic drop in success for passing in snow, pass plays saw a 10% reduction for snow inputs and running plays conversely increased by 10%. Rain resulted in a 5% increase in run plays and 5% decrease in pass plays. These weights can be adjusted by the user or coaches based on personnel decisions, toleration for aggression/risk or any other factors. The game plan is an interactive, multi layered treemap just like the original general game plan. This transition from general to weather gameplan can be seen as an illustrated form in Appendix B.

For sports betting, in our linear regression models for games between home and away teams, humidity consistently stood out as the most influential weather factor across all three model types—total passing yards, total rushing yards, and total points. This was evident as humidity consistently displayed the largest coefficient magnitude among wind, humidity, and temperature.

A final interesting observation we encountered for our Fantasy Football study was that as temperature increases, yards per attempt (YPA) for rushing plays decreases and the YPA for passing

plays increases. This suggests that managers should be playing more pass catchers in their Fantasy lineup on warmer days and more running backs on colder days. Appendix G demonstrates actual YPA scores versus predicted score obtained from the linear regression model.

Conclusions & Discussion

The NFL is as much a business as it is a professional sports league. Its reliance on analytics has increased over the past couple of years and continues to grow. Our analysis of how weather impacts performance can be another significant analytics tool that can be utilized in the league and the communities around it. Through random forests, train-test splits, and k-fold cross validation, our three models generate useful insight for the three platforms that would capitalize most from it: NFL team organizations, sportsbooks, and Fantasy Football consumers. The main takeaway from our analysis is that wind and precipitation (rain or snow) are the central conditions that have the most impact on performance, specifically rushing and passing, while humidity was found to have little correlation in Fantasy Football but was stood out as the most influential weather factor when it comes to sports betting. While it is easier for sports betters and Fantasy Football managers to utilize this information and apply it to their bets and games with a click of a button, game planning with this knowledge might be more difficult to implement in real-world scenarios as it needs organizational buy-in and coach persuasion.

Even though our analysis of climate data yielded insightful results, we came across a few limitations along the way. One of the limitations that we came across early on is the issue of missing data. In our dataset, we were missing approximately 30% of weather, humidity, temperature, and wind variables. Most of this missing data came from indoor stadiums, which we filled with average indoor dome conditions. For outdoor stadiums, we filled missing values with the average values for 23 years of games played at the specific stadiums. Snow and rain variables were also binary in nature. Future iterations of the data could be improved by better precipitation tracking.

Another limitation that has been at the center of debate in football analytics the past few years is the concept of intangibles. Our model, like other football analytics models, does not factor in intangibles such as injuries, momentum, morale, and exhaustion. During a game when a team is making a comeback or gaining momentum to take the lead when the score is close, confidence may build as a team executes their strategy on a late drive. These factors are not considered when using analytics to help make decisions. The model may tell coaches and decision makers the ideal decision based on data, but momentum, the atmosphere of the game or coaching expertise may say otherwise. Having the instinctual feel of the game is difficult to measure and our models do not consider these variables when generating our outcomes.

Lastly, the META in football is rapidly changing. As recently as 10 years ago, teams believed in a more run heavy offensive attack. In the modern NFL, we understand the value of passing and the tradeoffs made when a run play is called. The difference in eras and how offensive and defensive football has evolved was not captured by our model. Future versions may be better served with multiple models or an adjustment for era metric.

Nonetheless, we learned many valuable lessons from our analysis and are excited to see how our work performs against future data. All team members contributed equally to this project and have walked away with a better understanding of modeling, the NFL, weather impact on performance and illustrating our findings through visualization.

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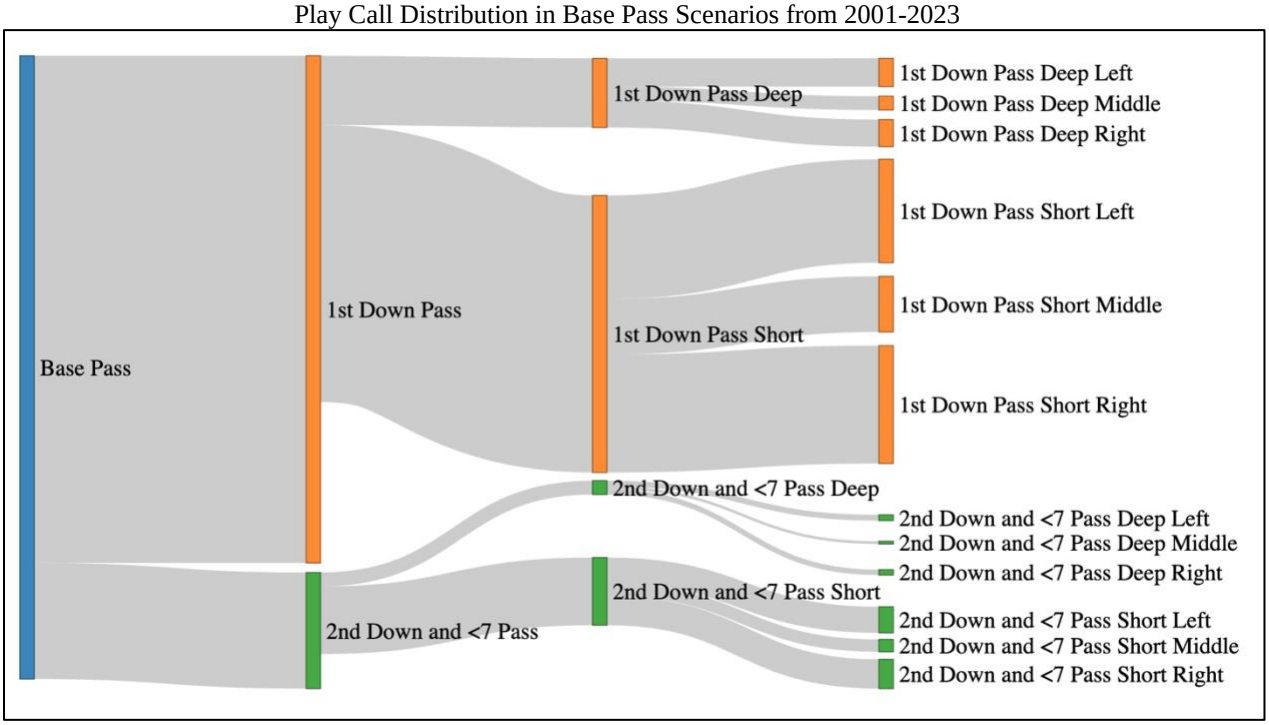
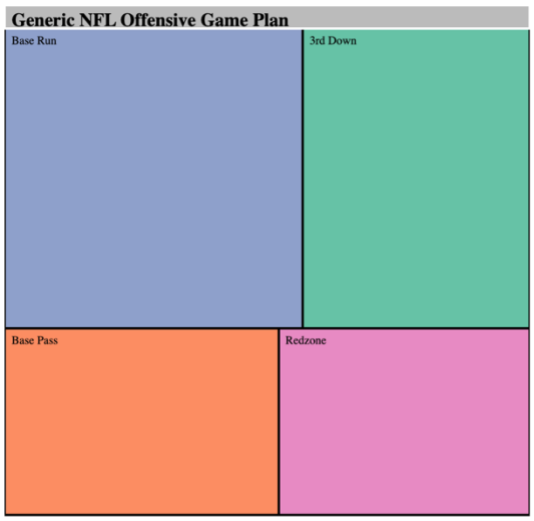
Appendix A

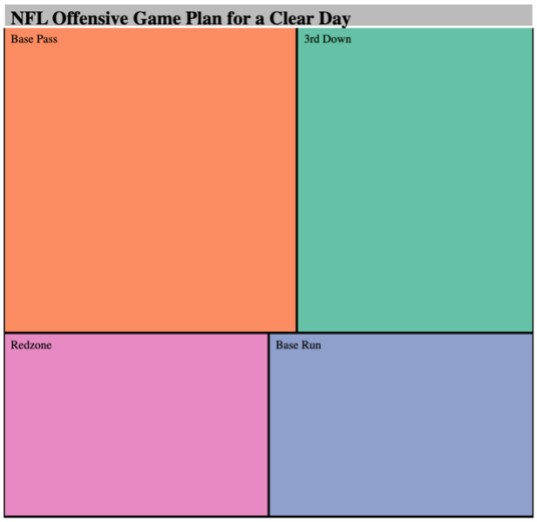
Play Call Distribution in Base Pass Scenarios from 2001-2023

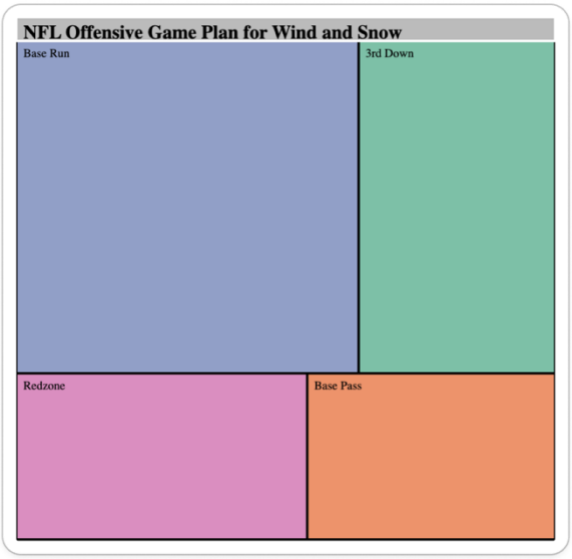


Appendix B

1). General Offensive Game Plan Based on Plays from 2001-2023
2). Gameplan Adjusted for Clear Day with Increased Passing
3). Gameplan Adjusted for Snowy and Windy Game with Increase in Rushing

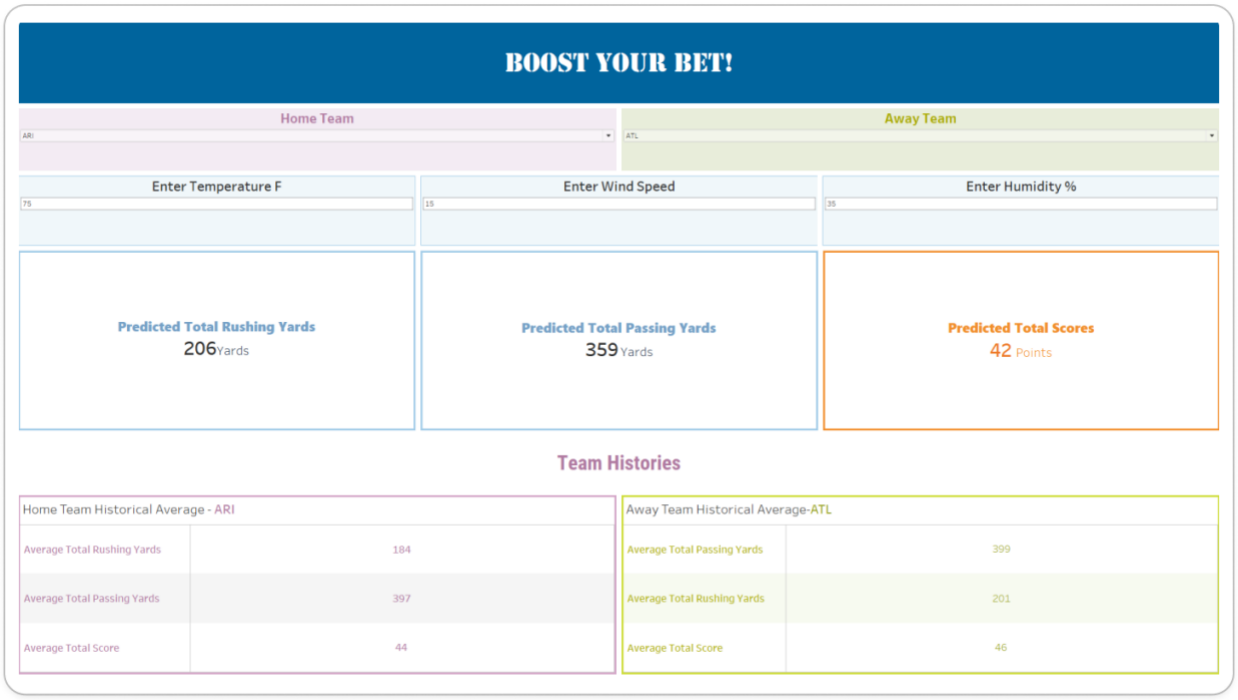






Appendix C

Sample of the UI of our NFL Statistical Player Performance Predictor

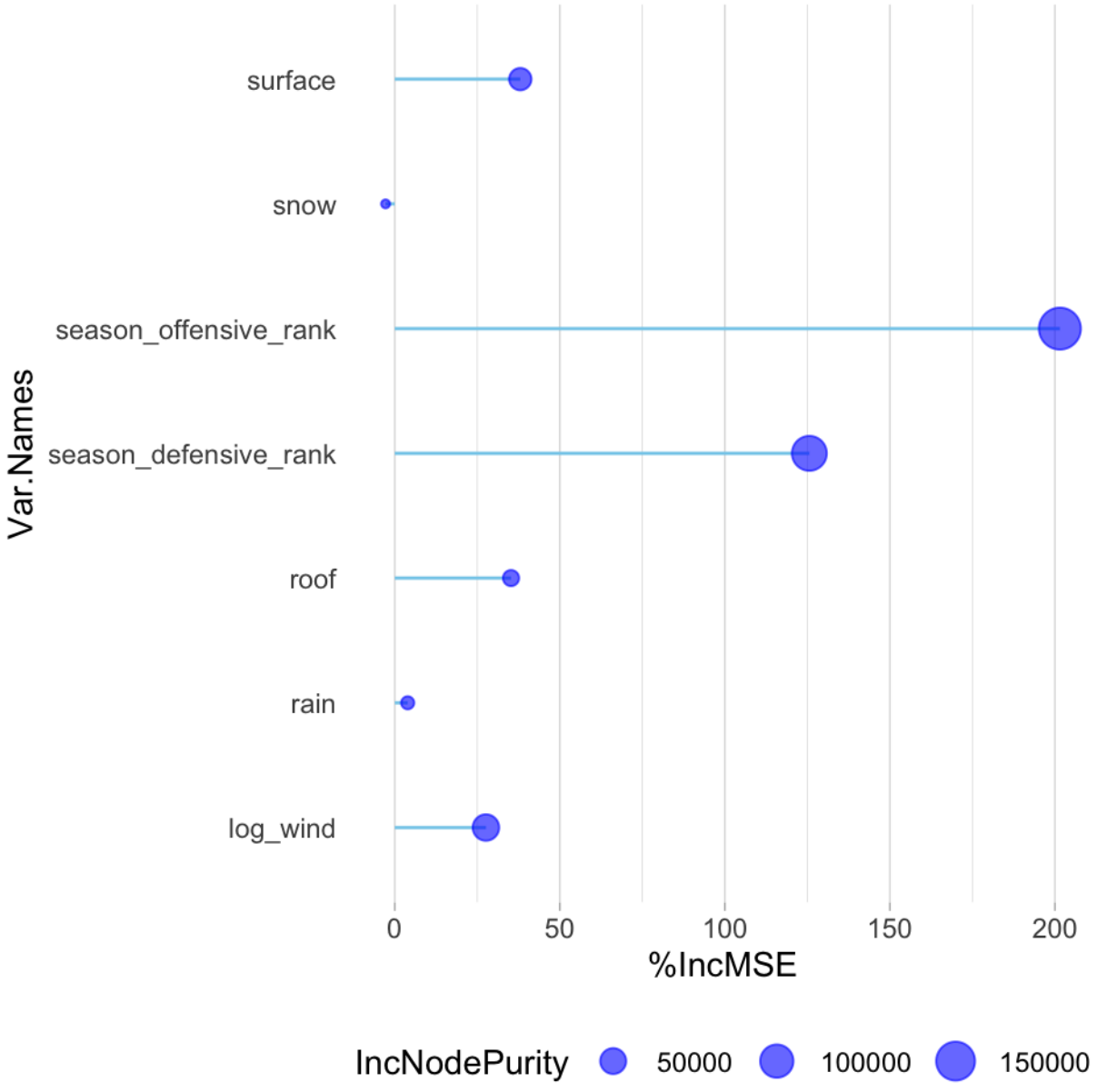


Appendix D

Sample of the UI of our Fantasy Football Visualization

Appendix E

Variance Chart for Random Forest Variables

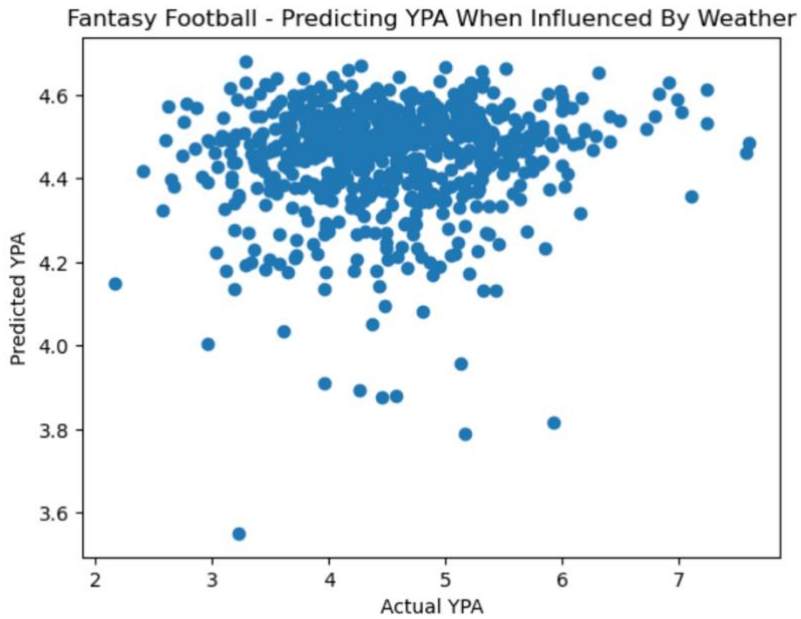


Appendix F

NFL Success Rate by Play Type and Weather (2001-2023)

Play Type	Normal Plays	More Wind	Less Wind	Rain	No Rain	Snow	No Snow
Pass	44.2%	42.1%	45.8%	41.0%	44.6%	38.8%	44.5%
Run	40.4%	40.2%	40.6%	40.7%	40.4%	40.9%	40.4%

Appendix G


Fantasy Football - Predicting YPA When Influenced By Weather
Predicted YPA
4.6
4.4
4.2
4.0
3.8
3.6
2
3
4
5
6
7
Actual YPA